MPA-RES 26/27-Z

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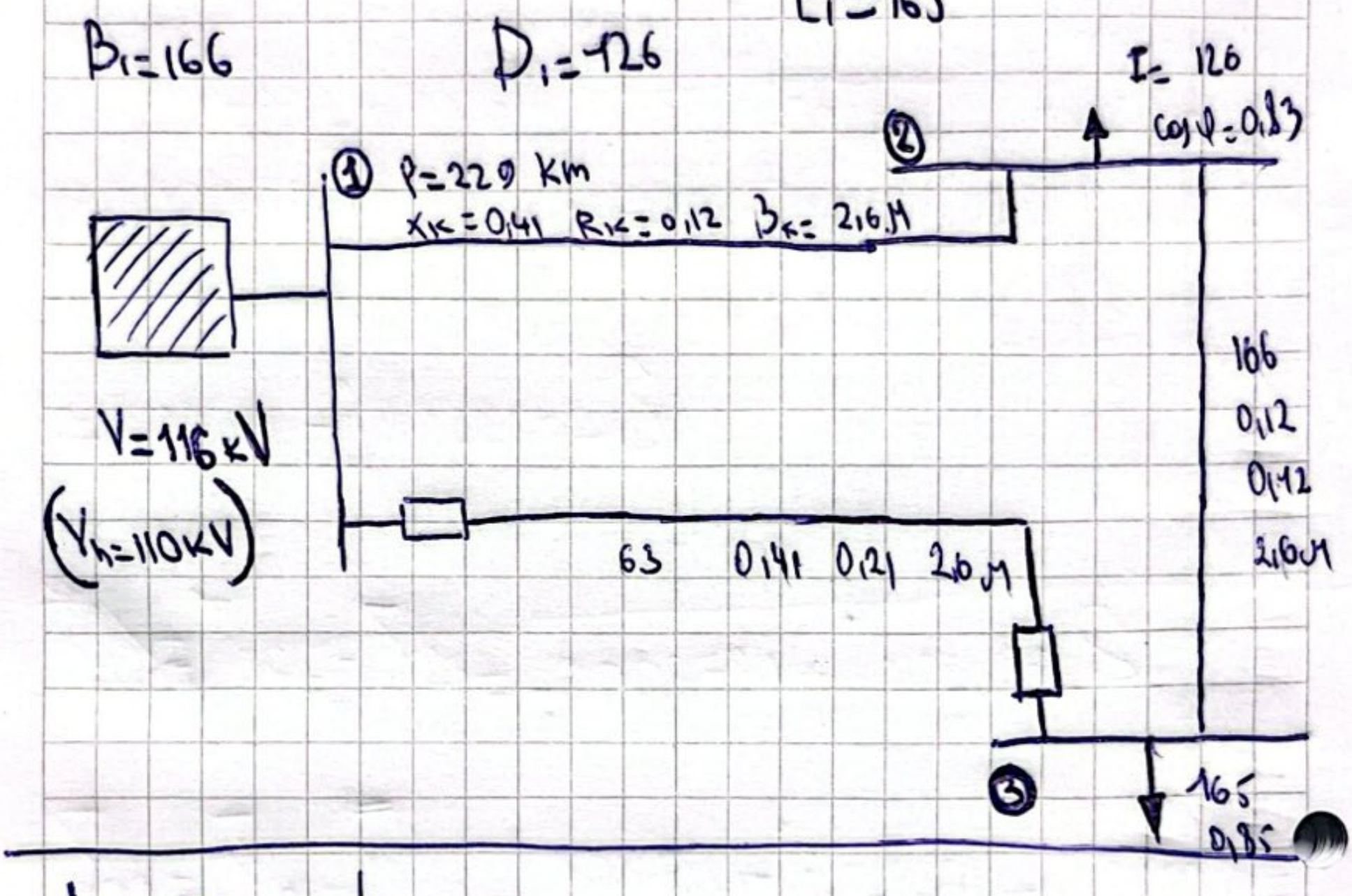
Individual Values:

$A_1 = 229$ $C_1 = 63$ $E_1 = 165$

$B_1 = 166$ $D_1 = 126$



Line parameters: [1-2]

$\bar{Z}_{12} = l(R_k + jX_k) = 229(0{,}12 + j0{,}41) = 27{,}48 + j93{,}89\,\Omega$

$\bar{Y}_{12} = 1/\bar{Z}_{12} = 0{,}002871 - j0{,}009810\,S$

$\bar{Y}_{120} = j(2{,}6 \times 10^{-6} \times 229)/2 = j0{,}000298\,S$

[2-3]

$\bar{Z}_{23} = (0{,}12 + j0{,}42) \times 166 = 19{,}92 + j69{,}72\,\Omega$

$\bar{Y}_{23} = 1/\bar{Z}_{23} = j0{,}003789 - j0{,}013261\,S$

$\bar{Y}_{230} = j(2{,}6 \times 10^{-6} \times 166)/2 = j0{,}000216\,S$

[1-3]

$\bar{Z}_{13} = (0{,}21 + j0{,}41) \times 63 = 13{,}23 + j25{,}83$

$\bar{Y}_{13} = 1/\bar{Z}_{13} = 0{,}01570 - j0{,}03067\,S$

$\bar{Y}_{130} = j(2{,}6 \times 10^{-6} \times 63)/2 = j0{,}0000819\,S$

[Load Currents:]

$-\bar{I}_2 = 126\angle -\arccos(0{,}83) = 126\angle -33{,}90° = -104{,}58 + j70{,}28\,A$

$-\bar{I}_3 = 165\angle -\arccos(0{,}85) = 165\angle -31{,}79° = -140{,}25 + j86{,}91\,A$

But they will be with minus sign, because they are injected currents to the Y-matrix.

Case a) Radial configuration (switches off)

In the radial configuration, Line 1-3 is physically disconnected

Then, Y-bus matrix will be:

$\bar{Y}_{22} = Y_{12} + Y_{23} + \bar{Y}_{120} + \bar{Y}_{230} = (0{,}002871 - j0{,}009810) + (0{,}003789 - j0{,}013261) + (j0{,}000298) + j0{,}000216 =$

$= 0{,}00666 - j0{,}022557\,S$

$\bar{Y}_{32} = -\bar{Y}_{23} = -0{,}003789 + j0{,}013261\,S$

$\bar{Y}_{33} = \bar{Y}_{23} + \bar{Y}_{230} = (0{,}003789 - j0{,}013261) + (j0{,}000216) =$

$= 0{,}003789 - j0{,}013045\,S$

$$\begin{bmatrix} \bar{Y}_{22} & \bar{Y}_{23} \\ \bar{Y}_{32} & \bar{Y}_{33} \end{bmatrix} \begin{bmatrix} \bar{V}_2 \\ \bar{V}_3 \end{bmatrix} = \begin{bmatrix} \bar{I}_2 + Y_{12}V_{1f} \\ \bar{I}_3 + \overset{0}{\cancel{Y_{31}V_{1f}}} \end{bmatrix}$$

$V_{1f} = \frac{116000}{\sqrt{3}} \angle 0° = 66972{,}6 + j0\,V$

$$\begin{bmatrix} 0{,}00666 - j0{,}022557 & 0{,}003789 - j0{,}013261 \\ -0{,}003789 + j0{,}013261 & 0{,}003789 - j0{,}013045 \end{bmatrix} \begin{bmatrix} \bar{V}_2 \\ \bar{V}_3 \end{bmatrix} =$$

$$= \begin{bmatrix} -104{,}58 + j70{,}28 + (0{,}002871 - j0{,}00981) \times 66972{,}6 \\ -140{,}25 + j86{,}91 \end{bmatrix}$$

$\Downarrow$

$$\begin{bmatrix} 0{,}00666 - j0{,}022557 & -0{,}003789 + j0{,}013261 \\ -0{,}003789 + j0{,}013261 & 0{,}003789 - j0{,}013045 \end{bmatrix} \begin{bmatrix} \bar{V}_2 \\ \bar{V}_3 \end{bmatrix} =$$

$$= \begin{bmatrix} 87{,}72 - j586{,}75\,A \\ -140{,}25 + j86{,}91\,A \end{bmatrix}$$

For finding Voltages we have to use inverse matrix of $\bar{Y}$-bus

inverse matrix:

$$\begin{bmatrix} \bar{Y}_{22} & \bar{Y}_{23} \\ \bar{Y}_{32} & \bar{Y}_{33} \end{bmatrix}^{-1} = \frac{1}{\bar{Y}_{22}\bar{Y}_{33} - \bar{Y}_{23}\bar{Y}_{32}} \times \begin{pmatrix} \bar{Y}_{33} & -\bar{Y}_{32} \\ -\bar{Y}_{23} & \bar{Y}_{22} \end{pmatrix}$$

It will be:

$$\frac{1}{(0{,}00666 - j0{,}022557) \times (0{,}003789 - j0{,}013045) - (0{,}003789 - j0{,}013261) \times (-0{,}003789 + j0{,}013261)} \times \begin{pmatrix} 0{,}003789 - j0{,}013045 & 0{,}003789 - j0{,}013261 \\ 0{,}003789 - j0{,}013261 & 0{,}00666 - j0{,}022557 \end{pmatrix}$$

$$[\bar{Y}]^{-1} = \begin{bmatrix} 31{,}69 + j100{,}15 & 32{,}62 + j101{,}53 \\ 32{,}62 + j101{,}53 & 54{,}10 + j173{,}63 \end{bmatrix} \Omega$$

$[\bar{V}] = [\bar{Y}]^{-1} [\bar{I}]$

$V_{2f} = (31{,}69 + j100{,}15)(87{,}72 - j586{,}75) + (32{,}62 + j101{,}53)(-140{,}25 + j86{,}91) = 48136 - j212206\,V$

$$\begin{bmatrix} V_{2f} = 48136 - j21206\,V = 52{,}6\angle -23{,}77°\,kV \\ V_2 = 91{,}1\angle -23{,}76°\,kV \end{bmatrix} \begin{matrix} L\text{-}N \\ L\text{-}L \end{matrix}$$

$V_{3f} = (62432 - j10230) + (-22681 - j19648) = 39751 - j29878\,V$

$$\begin{bmatrix} V_{3f} = 49{,}73\angle -36{,}93°\,kV \\ V_3 = 86{,}1\angle -36{,}93°\,kV \end{bmatrix} \begin{matrix} L\text{-}N \\ L\text{-}L \end{matrix}$$

b) Ring Configuration (Switches are on)

In this case $\bar{Y}_{33}$ will be changed:

$\bar{Y}_{33} = \bar{Y}_{23} + \bar{Y}_{230} + \bar{Y}_{13} + \bar{Y}_{130} = 0{,}019497 - j0{,}043632\,S$

New Node-Voltage system will be:

$$\begin{bmatrix} \bar{Y}_{22} & \bar{Y}_{23} \\ \bar{Y}_{32} & \bar{Y}^{*}_{33} \end{bmatrix} \begin{bmatrix} \bar{V}_2 \\ \bar{V}_3 \end{bmatrix} = \begin{bmatrix} \bar{I}_2 + \bar{Y}_{12}V_{1f} \\ \bar{I}_3 + \bar{Y}_{13}\bar{V}_{1f} \end{bmatrix}$$

Method will be same for finding voltages we will use the inverse form of updated $\bar{Y}$-bus matrix.

$$[\bar{Y}] = \begin{bmatrix} 0{,}00666 - j0{,}022557 & -0{,}003789 + j0{,}013261 \\ -0{,}003789 + j0{,}013261 & 0{,}019497 - j0{,}043632 \end{bmatrix}$$

$$[\bar{I}] = \begin{bmatrix} 87{,}72 - j586{,}75 \\ 911{,}76 - j1967{,}07 \end{bmatrix}$$

$$[\bar{Y}]^{-1} = \begin{bmatrix} 15{,}94 + j48{,}48 & 6{,}53 + j13{,}20 \\ 6{,}53 + j13{,}20 & 10{,}94 + j22{,}61 \end{bmatrix} \Omega$$

$[\bar{V}] = [\bar{Y}]^{-1} [\bar{I}]$

Then

$$\begin{bmatrix} V_{2f} = 61765 - j5894\,V = 62{,}045\angle -5{,}45°\,kV \\ V_2 = 107{,}5\angle -5{,}45°\,kV \end{bmatrix} \begin{matrix} L\text{-}N \\ L\text{-}L \end{matrix}$$

$$\begin{bmatrix} V_{3f} = 62771 - j3573\,V = 62{,}87\angle -3{,}25°\,kV \\ V_3 = 108{,}9\angle -3{,}25°\,kV \end{bmatrix} \begin{matrix} L\text{-}N \\ L\text{-}N \end{matrix}$$

[Tolerance:]

The range must be between 99kV to 121kV

Case A → $V_2 = 91{,}1$ kV and $V_3 = 86{,}1$ kV
They are not in the tolerance

Case B:
$V_2 = 107{,}5$ kV $V_3 = 108{,}9$ kV
They are in tolerance

CamScanner